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Referee #5 — Point-by-Point Assessment

Manuscript: POF26-AR-ICAFD2024-04570R — *Wall Shear Stress in Coronary Arteries and Grafts Using Computational Fluid Dynamics and Physics-Informed Neural Networks*, Physics of Fluids **Decision:** Major revision. Referees #1 and #2 accept; **Referee #5 is the gatekeeper. Revision due:** 08-Jul-2026 (clean “Article File” + marked-up “Marked-Up Manuscript”; alt-text Word document; highlight image).

This note assesses **whether each of Referee #5’s points is technically correct** — evaluated on the physics and verified against the current manuscript. Referee points 1 and 8 (and the editorial Figure-3 note) are being handled separately and are not covered here. Line numbers refer to the current manuscript source.

Bottom line

The review is careful and substantively correct on essentially every point. None of these items is wrong. Almost all are text edits or reframes. The only items needing new work are **one CFD validation case (Point 16)** and **optionally a proper grid-convergence index (Point 15)**. Two points are partly overstated, so we can push back (Points 11 and 13).

The items that most endanger us

- **Point 2 (Carreau–Yasuda mechanism) — “claimed fixed but not fixed.”** Our round-1 response letter describes the *correct* physics, but the manuscript still contains the original erroneous sentence **verbatim** — the corrected paragraph was never pasted in, and the referee noticed. Fix this first and acknowledge it in the response rather than re-claiming a fix.
- **Point 9a (Nannini misquote).** A verifiable factual error: we attribute $r = 0.988$ to “time-averaged WSS” when Nannini reported it for FFR-CT (a pressure ratio). Easy to lose credibility over.

Verdict table

#	Referee point	Correct?	Fix effort
2	Carreau–Yasuda “sensitivity” mechanism inverts the asymptotics (p.21)	Yes — fully (claimed fixed, still wrong)	Text reframe
3	“Protective low flow” reverses atherosclerosis physiology (p.25)	Yes — fully (contradicts our own Introduction)	Trivial
4	Rheology $\neq$ coagulation / NF- κ B / apoptosis (p.25)	Yes (verb softening)	Trivial
5	Bifurcation “stagnation point amplifies WSS” (p.18)	Yes (also fix Introduction)	Reframe
6	Spatial IQR described as “fluctuating” (temporal) (p.21)	Yes (one word)	Trivial
7	WSS definition (Eq 2.4) ambiguous vs the tensor Fluent reports	Yes	Trivial

#	Referee point	Correct?	Fix effort
9a	Nannini $r = 0.988$ is FFR-CT, not WSS	Yes — fully, decisive	Text
9b	Steady aortic-root inlet vs diastolic coronary timing	Yes — already partly acknowledged	One clause
9c	Single 40 dyn/cm^2 threshold for arteries <i>and</i> vein grafts	Yes (remedy “compartment-specific” is overstated)	Text
10	40 dyn/cm^2 derived and validated on the same 12 cases (circular)	Yes — fully	Reframe
11	Vessel-level statistics vs 12 patient geometries	Partly / overstated — acronyms <i>are</i> defined; mapping <i>is</i> recoverable	One sentence
12	“costs”/“expenses” = flow rates (translation artefact)	Yes — fully	Trivial
13	Correlations partly “built-in” (WSS $\leftrightarrow$ flow)	Partly / overstated — prescription right, rationale wrong	Temper wording
14	OSI > 0.1 , 15–25%, 10–15% unsubstantiated	Yes — OSI contradicts our own limitations	Text
15	%-difference $\neq$ GCI (Table 2)	Yes — fully	Reframe or real GCI
16	CFD not validated with our own solver	Yes (comparative design softens it)	One validation run

What “Point 16 — benchmark validation” actually requires (please read, CFD team)

A common misread: this is **not** asking us to validate the patient-specific models against another paper — that is impossible, because there is no ground-truth WSS for a real patient. The referee is asking us to **validate the solver/method, once, on a case that has a trusted answer**, run with *our* exact Fluent configuration, and then argue the validated method carries over to the patient geometries.

Distinction he is invoking: - **Verification** (what we have): mesh independence, residual convergence, prism layers $\rightarrow$ “are we solving the equations right?” - **Validation** (what is missing): comparing our solver output to an independent benchmark/experiment for a case in *our* configuration $\rightarrow$ “are we solving the right equations?”

Citing that other groups validated *their* solvers on the FDA nozzle / Sonmez / Kung does **not** transfer — validation is configuration-specific (solver version, laminar treatment, near-wall resolution, Carreau–Yasuda parameters, boundary conditions). Our round-1 rebuttal answered by citing third-party validation, which is exactly what he criticised, so this point is currently **un-addressed**.

Options, cheapest first: 1. **Analytic straight-tube check (recommended, ~hours).** Run a straight rigid tube in the *same* Fluent setup at a representative flow and show the computed wall WSS matches the analytic fully-developed Poiseuille value, $\tau = 4\mu Q/\pi R^3 = 4\mu U_{\text{mean}}/R$, reporting the % error. This is a clean closed-form check for the **Newtonian** solver and directly tests near-wall WSS resolution — the quantity the whole paper depends on. Note: a shear-thinning Carreau–Yasuda fluid has **no** simple closed-form Poiseuille WSS (the wall stress is set by the axial pressure gradient, $\tau_w = (R/2)(-dp/dx)$, and the flow-rate $\leftrightarrow$ wall-stress relation requires the Rabinowitsch–Mooney inversion, not $\tau = 4\mu Q/\pi R^3$). Validate the Carreau–Yasuda solver either against that semi-analytic pipe solution or via the experimental benchmark in Option 2. 2. **Reproduce a published experimental benchmark.** Re-run the FDA nozzle (or Sonmez’s 3D-printed stenosis, or the Kung/Kawasaki in-vitro case) with *our* solver settings/rheology and compare pressure drop / velocity / WSS quantitatively. More effort, strongest answer.

Mitigation we can also lean on: the paper’s headline claims are *relative* (diseased $>$ healthy WSS; Carreau–Yasuda more sensitive than Newtonian). Relative trends are robust to absolute calibration error, which lowers the bar — state this — but it does not remove the need for at least one quantitative own-solver case, because Carreau–Yasuda near-wall accuracy still hinges on validated WSS.

Detailed notes per point

Point 2 — Carreau–Yasuda high-shear mechanism is backwards (p.21)

$\mu_{\infty} = 0.0035 \text{ Pa}\cdot\text{s}$ **equals** the Newtonian viscosity, so at high shear the two models **converge** — they cannot be “amplified” there. Table 8 proves it (high-WSS zones: Graft 3(D) 147.8 vs 146.4; D2 LCA 83.7 vs 80.8). Also $\tau = \mu(\dot{\gamma})\cdot\dot{\gamma}$ grows **sub-linearly** under shear-thinning, so “increases non-linearly” (implying amplification) is wrong. Replace the passage with the response-letter language: convergence to μ_{∞} at high shear; the modest Carreau–Yasuda advantage comes from **low/intermediate-shear** regions where μ rises toward μ_0 . State the “more sensitive” claim cautiously — the advantage in Table 8 is only $\sim 1\text{--}3.5\%$. Also remove an internal inconsistency: the methods use the same 100 s^{-1} threshold to justify *Newtonian* modelling in large vessels.

Point 3 — “Protective low flow” (p.25, trivial)

The text says low flow is “protective (reduced oxidative stress)”; the Introduction correctly says low WSS *promotes* plaque. Low/oscillatory WSS is **pro-atherogenic**. Delete or reverse the “protective” clause and reframe the negative correlation as a statistical co-variation, not a biological mechanism. (The adjacent “laminar profile / minimises dissipation” clause is fine as pure fluid mechanics.)

Point 4 — Rheology $\neq$ coagulation (p.25, trivial)

A generalised-Newtonian viscosity law has no platelet/clotting or NF- κ B/apoptosis state variable. Replace “represents / reflect / faithfully reproduces” with “is consistent with / associated with”; drop or hard-qualify “hypercoagulation.” Keep only: altered WSS is a known mechanotransduction stimulus *associated with* endothelial signalling.

Point 5 — Bifurcation stagnation point (p.18, reframe)

At a true stagnation point velocity and WSS are **zero**; peak WSS is on the flow-divider **shoulders**. Re-attribute the acceleration / thin-boundary-layer mechanism to the shoulders just downstream of the apex. **Also fix the Introduction**, which currently puts high “WSS at the outer wall” of bifurcations — the outer wall is the classic **low/oscillatory**, plaque-initiation site. Add that low-WSS outer-wall counterpart.

Point 6 — IQR is spatial, not temporal (p.21, one word)

IQR over a single steady field is **spatial dispersion**. “Fluctuating shear stress” is the standard term for *temporal* oscillation (OSI), which a steady solve cannot produce — we even concede this in our own Limitations. Replace “fluctuating shear stress” with “spatially heterogeneous / large spatial dispersion of WSS.” Concede cleanly; don’t argue “spatial alternation = fluctuation.”

Point 7 — WSS definition (Eq 2.4, trivial)

Eq 2.4 uses an ambiguous $\partial v / \partial n$ with “v” undefined. WSS is the tangential projection of the viscous traction, $(I - n \otimes n) : (\tau \cdot n)$ — the quantity ANSYS Fluent reports as ground truth. Define the wall-tangential velocity gradient explicitly in Eq 2.4 and add one sentence stating that WSS is the tangential projection of the viscous traction (the Fluent quantity), of which the scalar form is the exact reduction at a Newtonian no-slip wall (and extends to variable viscosity).

Point 9 — Nannini citation + boundary conditions (text)

- **9a (decisive, must fix):** Nannini’s $r = 0.988$ is for **FFR-CT (a pressure ratio)**, not WSS — verified against the source. The Limitations wrongly states it as “time-averaged WSS”; the Methods says “haemodynamic indices.” Correct the Limitations to FFR-CT, reconcile the two passages, and either cite WSS-specific steady-vs-pulsatile evidence or soften to a limitation. (A pressure-drop index and a near-wall gradient quantity do not share sensitivity to unsteadiness.)
- **9b:** add one clause on diastole-dominant, out-of-phase coronary timing and ostial impingement. Already partly acknowledged; no new simulation strictly required.

- **9c:** justify the single 40 threshold across arteries and vein grafts as a *relative discriminator* (consistently applied across groups) and note vein-graft arterialization. The referee's "compartment-specific thresholds" remedy is overstated — a common threshold is defensible for a comparative study.

Point 10 — Threshold circularity (reframe)

40 dyn/cm² was chosen because it separated *these 12 cases*, then "validated" on the same 12 cases — that is circular, and 40 is not any of our own cited cutoffs (Introduction: >25 high, >50 rupture). Reframe as an **exploratory, sample-derived operating point**, acknowledge the in-sample circularity in the Limitations (currently absent), and reconcile with the 25/50 bands. (Feasible bonus: the two datasets give a natural train-on-one / test-on-other external check.)

Point 11 — Vessel vs patient (PARTLY OVERSTATED — we can push back)

Real part: the statistics are vessel-level (8 HCA / 15 PCA / 5 DG / 10 HG = 38) while the study covers 12 patient geometries. Add one mapping sentence or mini-table — it reconciles exactly from Table 1 (8 = H1–H4 RCA+LCA; 10 = 5 BG × 2 healthy grafts; 5 = 5 BG × Graft 3(D); 15 = BG/D native arteries). **But the referee is wrong that the acronyms are undefined** — HG/DG are defined at first use in the Statistical Analysis section, and HCA/PCA at first use in Results (rebut with this). Also ensure the leftover "nine cases" sentence stays removed/commented and is never reinstated.

Point 12 — "costs"/"expenses" → "flow rates" (trivial)

The correlation paragraphs call the flow-rate variables "costs"/"volume costs"/"expenses" (a translation artefact; the same variables are correctly called "flow rates" elsewhere). Replace in **those paragraphs only**. Do **not** global-replace "cost" — other occurrences correctly mean *computational* cost. Also fix "graphs" → "grafts."

Point 13 — Correlations "built-in" (PARTLY OVERSTATED)

Prescription is right (temper the interpretation), but the rationale is not fully: the inlet is a prescribed **velocity** (10 cm/s), not flow, and WSS↔inlet-flow is only 0.03 in our own matrix — so it is not the tautology claimed. Fixes: (i) state the WSS↔graft-flow link is a consistency check, not a discovery; (ii) **replace "Bernoulli effect"** — Bernoulli is a pressure/inviscid relation, wrong for a viscous WSS; (iii) soften "compensatory" (a mass-conservation closure, not biology); (iv) reframe the matrix's real added value as the geometry-selective pattern, the Newtonian-vs-Carreau–Yasuda $|r|$ attenuation, and inter-graft flow coupling.

Point 14 — Unsubstantiated numbers (text)

- **OSI > 0.1:** impossible from a steady solve, and it **contradicts our own Limitations** ("steady flow cannot capture OSI"). Delete it, or move it to future-work with explicit "not a result of this study."
- **15–25% bifurcation peak:** no derivation, no citation — remove or hedge + source.
- **10–15%:** actually **supportable** from our own correlation tables (Newtonian r 0.71/0.75 vs Carreau–Yasuda 0.60/0.66 $\approx$ 12–15%, restated as $\sim$ 15% later). Re-anchor to the in-paper tables instead of the external citation.

Point 15 — GCI (CFD side)

A %-difference is not a GCI (no observed order p , no Richardson extrapolation, no Roache safety-factor index). Two routes: **(A)** state plainly that only a qualitative peak-WSS mesh-sensitivity check was performed (this closes the referee's own "or" clause, low effort) — and remove the "well within the asymptotic convergence regime" wording, which asserts convergence without computing the order; or **(B)** compute a real Celik/ASME GCI — the three-mesh data already exist, and **4 of the 5 cases come out clean** (GCI below the reported raw %), so a real table would mostly *help* us. Caveat: the H1 LCA case is under-converged for peak WSS (true GCI $\sim$ 13%), so either refine it or report it honestly.

Point 16 — Own-solver validation (CFD side)

See the dedicated section above. Verification (mesh) $\neq$ validation. We need one quantitative case run with our own solver setup; the round-1 rebuttal cited third-party validation, which is exactly what the referee objected to.

Suggested division of work

- **CFD side:** Point 15 (GCI), Point 16 (one validation run), Point 9b (boundary-condition clause).
- **Writing (any author):** Points 2, 3, 4, 5, 6, 7, 9a, 9c, 10, 11, 12, 13, 14 — almost all text edits/reframes.

Effort summary

- **1 bounded analysis:** Point 16 (a few hours for the straight-tube check); optionally Point 15 (real GCI).
- **~12 text edits/reframes**, several trivial.
- **Highest priority:** the “claimed-fixed-but-not” item (Point 2) and the factual Nannini misquote (Point 9a) — these are the ones that cost credibility.

Source for Point 9a: Nannini et al. 2024, Computer Methods and Programs in Biomedicine 257:108415 (PubMed 39270532) — $r = 0.988$ reported for steady-vs-pulsatile FFR-CT.